LESSON 2.9

Describe Data by Mean Absolute Deviation

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the mean absolute deviation (MAD) to describe how spread out data is.

WHY THIS MATTERS The forecast shows the expected high temperatures for a week. The mean high temperature for the week is 79°F. How can you describe the variation in high temperatures over the week?

REMEMBER

Describe Data by Mean Absolute Deviation

Mean Absolute Deviation (MAD). Formula: MAD = (Sum of distances from mean) ÷ (Total count) The mean absolute deviation, or MAD, is the average distance of each value from the mean. It tells you how spread out the data is: a small MAD means the data is close together, and a large MAD means it is spread out.

1. Calculate the mean of the data set.
2. Find the distance (absolute difference |value − mean|) for each data point.
3. Average the distances: Add all distances and divide by the count

Words: **Mean Absolute Deviation *(Desviación media absoluta)*** **Deviation *(Desviación)*** **Absolute Value *(Valor absoluto)*** **Spread *(Dispersión)*** **Data distribution *(Distribución de datos)***

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Two soccer goalkeepers track goals allowed per game over 6 games. Keeper A: 1, 2, 1, 3, 2, 1 (mean = 1.67, MAD ≈ 0.67). Keeper B: 0, 4, 1, 3, 0, 2 (mean = 1.67, MAD ≈ 1.33). The coach needs to pick a starter for the championship game. What does a smaller MAD tell you?

(A) The values stay closer to the mean — the keeper is more consistent

(B) The keeper allowed fewer goals overall

(C) The keeper played fewer games

(D) The mean is wrong

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Find the mean of 10, 14, 8, 12 and 16.

(A) 10
(B) 12
(C) 14
(D) 60

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

3 **Circle the letter of the best answer. Show how you know.**

Four friends have 6, 9, 4 and 5 stickers. If they share them equally, how many does each get?

(A) 4
(B) 5
(C) 6
(D) 24

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A student's first three quiz scores are 82, 90 and 86. What score on a fourth quiz gives a mean of 87?

(A) 86
(B) 88
(C) 90
(D) 92

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

5 **Circle the letter of the best answer. Show how you know.**

The mean is called the balance point because…

(A) it is always the middle value in order

(B) the amounts above it and below it balance out

(C) it is the most common value

(D) it is always the largest value

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Two soccer goalkeepers track goals allowed per game over 6 games. Keeper A: 1, 2, 1, 3, 2, 1 (mean = 1.67, MAD ≈ 0.67). Keeper B: 0, 4, 1, 3, 0, 2 (mean = 1.67, MAD ≈ 1.33). The coach needs to pick a starter for the championship game. What is each keeper's mean goals allowed per game?

(A) about 1.67

(B) 2

(C) 6

(D) 10

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Player A's MAD is 2.4 points. If Player B's MAD is 7.2 points, what does that tell you about their consistency?

SENTENCE STARTER Player A has a ___ MAD, which means their scores are ___ to the mean. Player B has a ___ MAD, which means their scores are ___ from the mean. The more consistent player is ___ because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Each player's points are listed. Match each player's data set to its correct MAD. (Hint: find the mean first, then average the distances.)

_____ Bench: 8, 10, 12, 14

_____ Rookie: 4, 4, 4, 4

_____ Starter: 5, 15, 5, 15

(A) MAD = 2 points

(B) MAD = 0 points

(C) MAD = 5 points

2 **Write the name of the correct group on each line.**

Coach is sorting players by consistency. Drag each player's MAD into the bucket that describes their scoring.

Groups: More consistent (small MAD) | Less consistent (large MAD)

_____ Player with MAD = 1 point

_____ Player with MAD = 0 points

_____ Player with MAD = 9 points

_____ Player with MAD = 12 points

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Two soccer goalkeepers track goals allowed per game over 6 games. Keeper A: 1, 2, 1, 3, 2, 1 (mean = 1.67, MAD ≈ 0.67). Keeper B: 0, 4, 1, 3, 0, 2 (mean = 1.67, MAD ≈ 1.33). The coach needs to pick a starter for the championship game. Which keeper should the coach start, and why?

(A) Keeper B, because two shutouts is best

(B) Either one, because the means are equal

(C) Keeper A, because the smaller MAD (0.67) means more consistent performance

(D) Keeper B, because a larger MAD means more skill

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A guard scored 2, 4, 6, 8 points in four scrimmages, so the mean is 5 points. The distances from the mean are 3, 1, 1, 3. What is the mean absolute deviation (MAD)?

(A) 1

(B) 2

(C) 5

(D) 8

Put the data in order _____

WORK

5 **Circle the letter of the best answer. Show how you know.**

Two shooters average 11 points a game. Shooter A's MAD is 2 points and Shooter B's MAD is 6 points. Which shooter is MORE consistent?

(A) Shooter B, because a bigger MAD means a bigger average

(B) They are equally consistent because they have the same mean

(C) Shooter A, because a smaller MAD means scores stay closer to the average

(D) You cannot tell consistency from MAD

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — The Deviations That Vanished

1. **Data set** A center's rebounds: 3, 7, 11, 15, 19
2. **Find the mean** Sum = 55, Count = 5, Mean = 55 ÷ 5 = 11
3. **Find each deviation (value − mean)** −8, −4, 0, 4, 8
4. **Find the MAD** MAD = (−8 + −4 + 0 + 4 + 8) ÷ 5 = 0 ÷ 5 = 0

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Player A's MAD is 2.4 points. If Player B's MAD is 7.2 points, what does that tell you about their consistency?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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